Estimating Rare-Event Probabilities in Masked Diffusion Language Models with Multiple Exponential Tilts

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Abstract

Estimating the probability that a generative model produces a harmful output is difficult when the event is too rare to be reliably observed by direct sampling. We study this problem for masked diffusion language models (MDLMs), using complete denoising trajectories even when terminal-text probabilities are unavailable. The construction samples from the native decoder, but its rollback-MH correction does not evaluate native transition probabilities because they cancel analytically. We introduce a consistent estimation pipeline based on multiple exponentially tilted path distributions. Each tilt makes event-relevant outputs easier to observe; self-normalized importance sampling recovers the event probability under the native model, and the corrected estimates are combined convexly. We instantiate the tilted-population step with Trajectory SMC, which proposes suffixes with the unchanged decoder and applies an exact Metropolis–Hastings correction. As a case study of the general framework, we evaluate rare toxicity events and include FK Steering as an alternative tilted sampler. The evaluation separates statistical accuracy, resampling genealogy, terminal-sample diversity, and computational cost.

1 Introduction

Safety evaluations of generative models often report whether a model can be induced to produce a harmful output. Deployment decisions, however, can also depend on a different quantity: the probability that such an output occurs under the model's ordinary generation process. More generally, the event may be any measurable failure defined on the generated trajectory, such as toxicity, a policy violation, an unsafe generated action, or an erroneous tool invocation. The same objective can be defined conditionally for a fixed prompt or context c; our experiments focus on unconditional generation. This probability supports risk measurement and comparison, but it is difficult to estimate when failures are rare. With direct Monte Carlo, the relative error of an event-rate estimate grows rapidly as the probability decreases, and many evaluation runs may contain no event at all.

We consider rare-event probability estimation for masked diffusion language models (MDLMs) [Sahoo et al., 2024]. An MDLM generates a completion through a sequence of stochastic reverse-denoising transitions. Although this native path law is tractable to simulate, the marginal probability of a terminal text is not generally available as a product over its final tokens. We therefore perform rare-event simulation on complete native decoder trajectories rather than requiring terminal likelihoods.

Our pipeline introduces several auxiliary distributions that exponentially upweight trajectories with large values of an event-informative terminal observable. A particle method produces a sample

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population for each tilt. We then reverse the tilt separately within each population using self-normalized importance sampling (SNIS), and combine the component estimates only after each ratio has been formed. The tilted populations are evaluation instruments: the estimand remains the event probability under the frozen, unmodified decoder.

The pipeline treats the method used to obtain a tilted population as a modular choice. We instantiate it with Trajectory SMC, whose mutation rolls a stored native trajectory back to a checkpoint and regenerates its future with the unchanged decoder. The resulting Metropolis–Hastings (MH) ratio depends only on the difference in the terminal observable. To test the pipeline's sampler interface, the experiments additionally instantiate it with FK Steering [Singhal et al., 2025], a recent particle method for diffusion steering. Both methods construct auxiliary tilted populations; neither defines a replacement decoding policy or changes the probability being estimated.

Our contributions are:

- a consistent multi-tilt estimation pipeline that corrects separately normalized tilted populations by SNIS and combines them to estimate a rare-event probability under the original MDLM; and
- Trajectory SMC, a path-space construction for MDLMs whose native-decoder suffix proposals make the decoder transition probabilities cancel analytically in the rollback-MH ratio. Its acceptance probability therefore depends only on the change in the terminal observable, without redefining the underlying decoding policy.

2 Relation to Prior Work

Masked diffusion language models. Masked diffusion language models generate text through a reverse process that progressively replaces masked positions with tokens. We build on the MDLM formulation of Sahoo et al. [Sahoo et al., 2024] and use its native reverse transition kernel without retraining the model. Whereas prior evaluations of MDLMs primarily study generation quality and sampling efficiency, our objective is to estimate the probability of rare events under the model's original path distribution; classifier-defined toxicity events provide the experimental instantiation studied here.

Particle methods for diffusion generation. Recent work uses interacting particle systems to improve or control diffusion-model generation at inference time. FK Steering constructs a Feynman–Kac particle system whose particles are weighted and resampled using sequence-level rewards evaluated through predicted terminal completions [Singhal et al., 2025]. Self-Rewarding SMC similarly maintains multiple MDLM trajectories, but derives its particle weights from trajectory-level model confidence [Luo et al., 2026]. These methods demonstrate that particle computation can guide diffusion trajectories without additional model training. To our knowledge, rare-event probability estimation for masked diffusion language models has not previously been studied. Unlike steering methods, our goal is to estimate the probability of a rare event under the model's original generation process.

Rare-event simulation and path-space SMC. Importance sampling and self-normalization are classical tools for estimating rare-event probabilities from samples drawn under an alternative distribution. Feynman–Kac particle systems extend this construction to sequences of intermediate targets, while resample–move methods introduce invariant Markov transitions to rejuvenate populations after resampling [Del Moral, 2004, Gilks and Berzuini, 2001]. Our work follows this statistical perspective: tilted particle populations are auxiliary proposals rather than the desired output distribution. Trajectory SMC uses native-decoder rollback proposals within this construction, while the subsequent SNIS correction targets the rare-event probability under the unchanged MDLM.

3 Rare-event probabilities on MDLM trajectories

Let $\tau = (x_T, x_{T-1}, \ldots, x_0)$ be a complete native denoising trajectory, where x_T is the initial masked state and x_0 is the terminal text. The frozen decoder defines

$$P(\tau) = p_T(x_T) \prod_{t=1}^{T} K_t(x_t, x_{t-1}). \tag{1}$$

We can sample from P by running the decoder, without evaluating the marginal probability of an intermediate state or terminal text. We write $P(\tau)$ for notational simplicity. For a fixed prompt or context c, the same construction applies conditionally by replacing $P(\tau)$ throughout with $P(\tau \mid c)$.

Let $\mathcal{A}$ be a rare set of terminal outputs. We aim to estimate

$$p = \mathbb{E}_P[\mathbb{I}\{X_0 \in \mathcal{A}\}]\,. \tag{2}$$

The target is always defined under the original decoder. Direct Monte Carlo averages this indicator over independent native trajectories, but becomes inefficient when most trajectories do not enter $\mathcal{A}$.

We assume access to a terminal observable $\phi : \mathcal{X} \to \mathbb{R}$ that carries information about the event. It need not equal the binary indicator; its role is to provide a graded signal that tends to be larger for outputs more closely associated with $\mathcal{A}$. For $\lambda \geq 0$, define

$$P_\lambda(\tau) = \frac{P(\tau)e^{\lambda\phi(x_0)}}{Z_\lambda}, \qquad Z_\lambda = \mathbb{E}_P[e^{\lambda\phi(X_0)}]. \tag{3}$$

An informative observable makes the rare-event region easier to visit. It affects sampling efficiency, not the estimand or the validity of the reverse correction, provided the required moments are finite.

4 Three-step multi-tilt SNIS pipeline

The pipeline separates the statistical estimator from the algorithm used to generate its proposal samples. It has three steps.

Step 1: SNIS with a generic proposal. Let Q be a proposal distribution on complete trajectories whose support contains that of P. If $\tau^1, \ldots, \tau^N$ are sampled from Q, define a reverse importance factor $r(\tau)$ proportional to $P(\tau)/Q(\tau)$. SNIS estimates the event probability by the ratio

$$\widehat{p}_Q = \frac{\sum_{n=1}^N r(\tau^n)\mathbb{I}\{x_0^n \in \mathcal{A}\}}{\sum_{n=1}^N r(\tau^n)}. \tag{4}$$

The numerator estimates how much proposal mass belongs to the event after correction toward P; the denominator restores the overall scale.

Step 2: Trajectory SMC. Direct sampling corresponds to $Q = P$ and rarely observes the event. We instead choose the tilted proposal $Q = P_\lambda$ from Eq. (3). When ϕ is informative, this proposal assigns more mass to trajectories ending near or inside the event. Sampling from P_λ is a separate module and can be carried out by different Monte Carlo methods. We use Trajectory SMC, a variation of the established SMC sampler framework, to obtain a particle approximation of the tilted proposal while retaining the native MDLM as the base path law. Its intermediate targets, particle weights, resampling step, and rollback mutation are specified and derived in Appendices A–C.

Step 3: average across several tilts. The best tilt is not known in advance and may depend on the model, event, observable, threshold, and particle budget. Weak tilts may expose too few rare events, whereas strong tilts can destabilize the reverse correction. Identifying one tilt through dedicated pilot runs would require additional expensive model evaluations. We therefore use a pre-specified tilt ladder $\Lambda = \{\lambda_1, \ldots, \lambda_J\}$ and combine all separately corrected estimates, rather than selecting one tilt from the evaluation data. In our experiments we use the uniform average

$$\widehat{p} = \frac{1}{J}\sum_{j=1}^J \widehat{p}_{\lambda_j}. \tag{5}$$

The uniform coefficients are fixed before evaluation and are not claimed to be optimal. They are a design choice for our experiments, not a requirement of the pipeline: more generally, the component estimates can be combined with nonnegative weights summing to one, and those weights may be optimized when a suitable selection procedure or pilot budget is available.

5 Experimental evaluation

Evaluation goals and metrics. We test the complete pipeline as a probability estimator rather than only its ability to discover high-scoring texts. The evaluation asks four separate questions. First, *statistical accuracy* is measured over independent complete runs using standard deviation, RMSE relative to a direct Monte Carlo reference, and the fraction of zero estimates. Second, *correction stability* is measured for every tilt through its component estimate and reverse-correction ESS. Third, we distinguish *resampling genealogy*, measured by surviving roots, root entropy, largest root family, and resampling counts, from *terminal sample diversity*, measured by exact distinct token sequences and duplicate-group size. Finally, *computational cost* uses wall time, decoder particle-steps, classifier evaluations, and peak accelerator memory. None of these diagnostics is treated as a surrogate for the others.

Model and evaluator. We estimate event probabilities for unconditional samples from the frozen MDLM-OWT model. Generation starts from 128 mask tokens and uses 32 stochastic DDPM/SUBS denoising steps. Let $S_{\mathrm{tox}}(x)$ be the positive-class probability from a frozen RoBERTa-based toxicity classifier applied to the first 100 generated MDLM tokens. For threshold a, the rare set is $\mathcal{A}_a = \{x : S_{\mathrm{tox}}(x) \geq a\}$. Motivated by classifier-derived rewards in prior diffusion-steering work, we instantiate the generic observable as $\phi(x) = \log S_{\mathrm{tox}}(x)$. Classifier padding and final-microbatch handling are fixed so that the observable is independent of other particles evaluated in the same batch. We study nested thresholds $a \in \{0.5, 0.9, 0.99, 0.999\}$, designate $a = 0.99$ as primary, and treat $a = 0.999$ as exploratory.

Reference and Trajectory SMC design. The direct reference contains 200,000 valid unique IID completions from the same native decoder. It contains 444 events at $a = 0.99$ and 26 at $a = 0.999$; the latter reference is consequently uncertain despite its large sample size. It is a Monte Carlo reference, not exact ground truth. Each SMC estimate uses four independently generated components with $\lambda \in \{0.25, 0.5, 0.75, 1.0\}$ and uniform coefficients. The final analysis contains 64 independent Trajectory SMC runs at both $N = 192$ and $N = 512$, pooling two disjoint 32-run cohorts where an earlier submission or runtime-matched cohort was available. Confirmatory seeds are disjoint from pilot and runtime-calibration seeds.

Trajectory SMC uses eight tempering levels, one mutation sweep per level, and rollback depths drawn uniformly from $\{4, 8, 16, 24, 32\}$, with systematic resampling at normalized ESS below 0.5.

Additional tilted sampler: FK Steering. To evaluate whether the estimation pipeline remains useful with a different tilted-population mechanism, we additionally use FK Steering, a recent Feynman–Kac particle method for diffusion models [Singhal et al., 2025]. This comparison contextualizes Trajectory SMC and tests the modular sampler interface; it is not the primary estimand and is not a comparison between replacement decoding policies. FK propagates particles with the native diffusion kernel and resamples them using intermediate reward potentials. At each checkpoint, our implementation scores four disposable one-shot samples from the predicted clean-token distribution; these guide resampling but never replace the natively propagated particle. The terminal potential scores the actual completion. We apply the same SNIS correction and uniform multi-tilt average to both samplers, using the same decoder, observable, thresholds, tilts, particle budgets, and independent-seed policy. Appendix E gives the FK implementation and its telescoping terminal potential. FK uses systematic resampling at the same ESS threshold and evaluates four one-shot predictions every four denoising steps. The final analysis contains 64 FK runs at $N = 512$ and 32 confirmatory FK runs at $N = 192$.

6 Results

The direct reference gives $p = 0.00222$ at the primary threshold, with binomial standard error 1.05×10^{-4} and Wilson 95% interval $[0.002023, 0.002436]$. It is a finite Monte Carlo reference rather than exact ground truth. Table 1 summarizes independent complete SMC estimates. No run returned a zero estimate at this threshold.

Probability-estimation accuracy. At equal $N = 512$, the observed RMSE is descriptively lower for Trajectory SMC than for FK: the run-level bootstrap ratio is 0.509, with 95% interval $[0.305, 1.027]$.

Table 1: Probability estimation at $a = 0.99$. Runtime is the sum over four independently generated tilted components. The direct reference is 0.00222.

Engine	N	runs	mean $\widehat{p}$	SD	RMSE	time (s)
FK Steering	192	32	0.002476	0.002096	0.002079	254
Trajectory SMC	192	64	0.002042	0.001365	0.001366	710
FK Steering	512	64	0.002470	0.001514	0.001523	678
Trajectory SMC	512	64	0.002187	0.000781	0.000776	1886

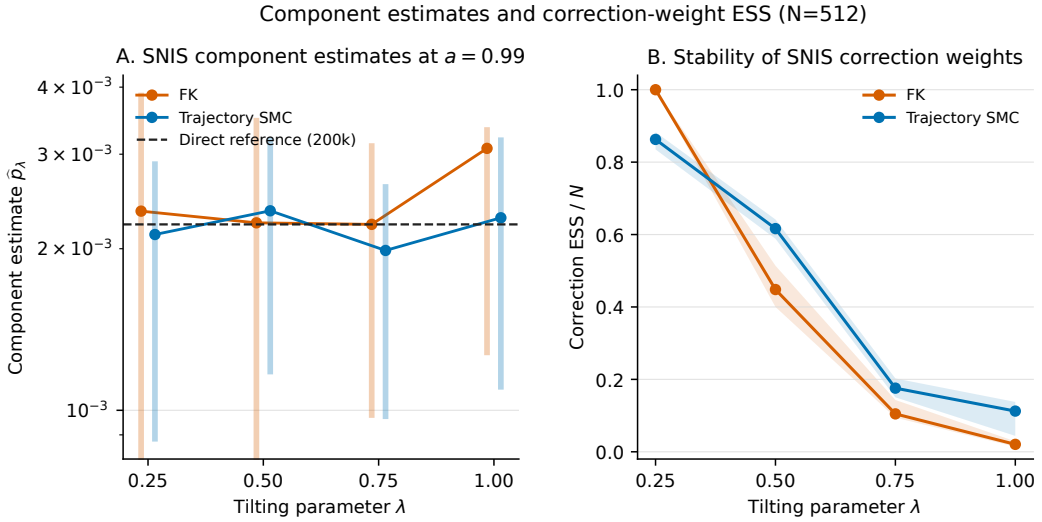


Figure 1: Component behavior across the tilt ladder at $a = 0.99$ and $N = 512$ (64 independent populations per point). Left: markers are mean component estimates and bars span the interquartile range; the dashed line is the direct reference. Right: markers and bands are the median and interquartile range of the SNIS correction-weight ESS.

Because the interval includes one, these runs do not establish a statistically resolved accuracy improvement. At the exploratory threshold $a = 0.999$, the direct reference contains only 26 events. The zero-estimate fractions are 0.203 for Trajectory SMC and 0.438 for FK at $N = 512$, which diagnoses event discovery but does not by itself establish accuracy.

Accuracy–cost trade-off. Trajectory SMC at $N = 512$ takes $2.78\times$ the median wall time of FK at the same particle count. The approximately runtime-matched comparison is therefore Trajectory SMC at $N = 192$ against FK at $N = 512$: their median runtimes are 710 and 678 seconds, and the RMSE ratio is 0.897 with bootstrap interval $[0.492, 1.939]$. The two methods spend computation differently. At $N = 512$, Trajectory SMC uses 340,704 decoder particle-steps and scores 22,528 classifier texts, whereas FK uses 65,536 decoder particle-steps and scores 61,440 texts; peak allocated accelerator memory is approximately 3 GB for both.

Component stability across tilts. Figure 1 shows why the pipeline does not rely on one preselected tilt. At $\lambda = 1$, the FK component has mean 0.00308, SD 0.00482, and median correction ESS/$N = 0.0206$. The corresponding Trajectory SMC component has mean 0.00228, SD 0.00149, and median correction ESS/$N = 0.1124$. Thus, in this experiment, native-suffix resample–move produces a more stable strongly tilted component. This is an empirical finite-particle observation, not a universal consequence of MH invariance. Uniformly averaging the four stored component estimates reconstructs every reported final estimate to machine precision, independently validating the aggregation pipeline.

Genealogy and terminal diversity. At $N = 512$ and $\lambda = 1$, the median fraction of surviving roots is 0.086 for FK and 0.270 for Trajectory SMC. The ordering reverses for exact terminal

sequence uniqueness: the medians are 1.000 and 0.936, respectively. Figure 2 reports all tilts. FK descendants can diverge during subsequent stochastic denoising even after early resampling; Trajectory SMC resamples complete paths, and only accepted suffix mutations can differentiate their copies. Genealogical survival therefore measures resampling dependence, whereas exact sequence uniqueness measures realized duplication. Neither is a surrogate for statistical accuracy or semantic diversity.

7 Limitations

Methodologically, the construction extends directly to a fixed conditional path law $P(\tau \mid c)$. Empirically, however, this study evaluates only classifier-defined events for one unconditional MDLM; prompt-conditioned generation has not been tested. Broader experiments are also needed across MDLM architectures, domains, event thresholds, and decoding configurations. Results for the present model and implementation do not establish universal superiority of either tilted-population engine.

The toxicity classifier is an imperfect proxy for harmfulness and may contain domain, linguistic, and demographic biases; rare-event estimation does not correct those measurement errors. We use one classifier-derived observable, and different observables or learned look-ahead functions may change the efficiency of the tilted populations. Our experimental observable is a bounded classifier score in $[0, 1]$. Near the upper end of its range, increasingly severe outputs can receive very similar scores, limiting the resolution with which progressively rarer events can be defined. In our stored samples, terminal scores concentrate below approximately 0.9997, and no outputs exceed 0.9999, so thresholds substantially beyond 0.9995 cannot be meaningfully evaluated in the current setting. This is a limitation of the experimental observable and benchmark, rather than of the path-space formulation itself.

The direct reference has uncertainty, especially at the most extreme threshold. Exact token-sequence uniqueness also does not measure semantic diversity.

Finally, uniform component weights avoid a separate pilot optimization but do not exploit systematic differences among tilts. Principled data-dependent weights would need to account for finite-sample bias as well as component variances and covariances, preferably without reusing the evaluation runs for selection. A finite-sample or asymptotic characterization of the full SMC–SNIS variance, together with covariance-aware weight selection, is left for future work. Rollback storage and native suffix regeneration also create a computational cost that may limit scaling to longer trajectories.

8 Conclusions

The proposed pipeline turns tilted trajectory sampling into an estimator of risk under the unchanged model. Its modular construction allows established steering methods and path-space SMC alternatives to be evaluated under a common estimand and correction. Trajectory SMC provides a theoretically transparent option: each mutation uses the original multi-step decoder and is exactly invariant for the current tilted path measure. In the evaluated setting, its estimates agree with the direct Monte Carlo reference within the observed run-to-run uncertainty, and its strongest tilted component is more stable, at increased decoder cost. FK remains a valid alternative with cheaper runtime and high terminal uniqueness. Rare-behavior discovery and rare-behavior probability estimation are distinct objectives: particle steering can expose rare failures while also serving as an auxiliary mechanism for estimating their probability under the unchanged native generative process.

References

Nicolas Chopin. Central limit theorem for sequential monte carlo methods and its application to bayesian inference. *The Annals of Statistics*, 32(6):2385–2411, 2004. doi: 10.1214/009053604000000698.

Pierre Del Moral. *Feynman–Kac Formulae: Genealogical and Interacting Particle Systems with Applications*. Springer, 2004.

Pierre Del Moral, Arnaud Doucet, and Ajay Jasra. Sequential monte carlo samplers. *Journal of the Royal Statistical Society: Series B*, 68(3):411–436, 2006. doi: 10.1111/j.1467-9868.2006.00553.x.

Walter R. Gilks and Carlo Berzuini. Following a moving target—monte carlo inference for dynamic bayesian models. *Journal of the Royal Statistical Society: Series B*, 63(1):127–146, 2001.

Ziwei Luo, Ziqi Jin, Lei Wang, Lidong Bing, and Thomas B. Schön. Self-rewarding sequential monte carlo for masked diffusion language models, 2026.

Subham Sekhar Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin T. Chiu, Alexander M. Rush, and Volodymyr Kuleshov. Simple and effective masked diffusion language models. In *Advances in Neural Information Processing Systems*, 2024.

Raghav Singhal, Zachary Horvitz, Ryan Teehan, Mengye Ren, Zhou Yu, Kathleen McKeown, and Rajesh Ranganath. A general framework for inference-time scaling and steering of diffusion models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 55810–55827, 2025.

Appendix

A Complete Trajectory SMC implementation

For the tilted proposal in Eq. (3), the reverse importance factor is

$$\frac{P(\tau)}{P_\lambda(\tau)} = Z_\lambda e^{-\lambda\phi(x_0)} \;\propto\; e^{-\lambda\phi(x_0)}. \tag{6}$$

Substitution into Eq. (4) gives, for direct samples from P_λ,

$$\widehat{p}_\lambda = \frac{\sum_{n=1}^N e^{-\lambda\phi(x_0^n)}\mathbb{I}\{x_0^n \in \mathcal{A}\}}{\sum_{n=1}^N e^{-\lambda\phi(x_0^n)}}. \tag{7}$$

The unknown normalizer cancels between the two sums. In the actual SMC implementation, the sampler instead returns a normalized weighted approximation $\{(\tau^n, W^n)\}_{n=1}^N$. The corresponding component estimator is

$$\widehat{p}_\lambda = \frac{\sum_{n=1}^N W^n e^{-\lambda\phi(x_0^n)}\mathbb{I}\{x_0^n \in \mathcal{A}\}}{\sum_{n=1}^N W^n e^{-\lambda\phi(x_0^n)}}. \tag{8}$$

This reduces to Eq. (7) when the SMC weights are uniform. For any normalized weight vector v, we define

$$\mathrm{ESS}(v) = \frac{1}{\sum_{n=1}^N (v^n)^2}. \tag{9}$$

Forward SMC ESS is computed from W, whereas reverse-correction ESS is computed after multiplying W^n by $e^{-\lambda\phi(x_0^n)}$ and normalizing.

For a fixed tilt, use the intermediate targets

$$P_{\eta_k}(\tau) \propto P(\tau)e^{\eta_k\phi(x_0)}, \qquad 0 = \eta_0 < \cdots < \eta_K = \lambda. \tag{10}$$

The rollback proposal at level k is accepted with probability

$$A_k(\tau, \tau') = \min\{1, e^{\eta_k[\phi(x_0')-\phi(x_0)]}\}. \tag{11}$$

Algorithm 1: Multi-tilt Trajectory SMC estimator

Input: native decoder P; rare set $\mathcal{A}$; observable ϕ; tilts $\Lambda = \{\lambda_1, \ldots, \lambda_J\}$; particle count N; tempering schedules; normalized ESS threshold κ; rollback law ρ; and M MH sweeps per level.

1. For each tilt $\lambda \in \Lambda$:

a. Draw N complete trajectories $\tau^1, \dots, \tau^N$ from P, store their visited states, and set $W^n = 1/N$.

b. For $k = 1, \dots, K$:

i. multiply each W^n by $e^{(\eta_k - \eta_{k-1})\phi(x_0^n)}$ and normalize;

ii. if $\mathrm{ESS}(W) < \kappa N$ and $k < K$, resample complete trajectories, stored states, and genealogy, then set $W^n = 1/N$;

iii. for each MH sweep and trajectory, draw $d \sim \rho$, roll back by d native steps, and regenerate the suffix with the unchanged decoder;

iv. accept the proposed trajectory using Eq. (11).

c. Without final resampling, compute $\widehat{p}_\lambda$ using Eq. (8).

2. Return $\widehat{p} = J^{-1} \sum_{\lambda \in \Lambda} \widehat{p}_\lambda$.

Resampling can remove root lineages permanently. Rollback-MH cannot restore a lost ancestor, but can allow duplicated descendants to develop different suffixes. Genealogical survival, terminal sequence diversity, correction ESS, statistical error, and cost therefore remain distinct diagnostics.

B Derivation of the tempered SMC weights

For one terminal tilt λ, define the unnormalized targets

$$\gamma_k(\tau) = P(\tau) e^{\eta_k \phi(x_0)}, \qquad k = 0, \dots, K, \tag{12}$$

with normalizers $Z_k = \int \gamma_k(\tau) d\tau$ and normalized targets $P_{\eta_k} = \gamma_k / Z_k$. Since $\eta_0 = 0$, $\gamma_0 = P$ and $Z_0 = 1$.

The generic annealed-SMC reweighting step multiplies the current weight by the ratio of consecutive unnormalized targets:

$$G_k(\tau) = \frac{\gamma_k(\tau)}{\gamma_{k-1}(\tau)} \tag{13}$$

$$= \frac{P(\tau) e^{\eta_k \phi(x_0)}}{P(\tau) e^{\eta_{k-1} \phi(x_0)}} = e^{(\eta_k - \eta_{k-1})\phi(x_0)}. \tag{14}$$

Thus

$$W_{k,i} = \frac{W_{k-1,i} G_k(\tau_i)}{\sum_{\ell=1}^N W_{k-1,\ell} G_k(\tau_\ell)}. \tag{15}$$

The same denominator estimates the normalizer increment:

$$\frac{\widehat{Z}_k}{\widehat{Z}_{k-1}} = \sum_{i=1}^N W_{k-1,i} G_k(\tau_i). \tag{16}$$

If resampling is triggered, the resampled population receives uniform weights. A subsequent Markov kernel invariant for P_{η_k} requires no additional importance correction; this is the role of rollback-MH.

C Detailed balance for native rollback-MH

Draw a rollback depth d from a fixed, state-independent law ρ. Restore the checkpoint visited d native steps before termination, retain the realized prefix, and regenerate the remaining suffix with the unchanged decoder. Write the current and proposed trajectories as $\tau = (u, v)$ and $\tau' = (u, v')$, where u is their shared prefix. Their native path probabilities factorize as

$$P(\tau) = P(u)P(v \mid u), \qquad P(\tau') = P(u)P(v' \mid u). \tag{17}$$

Because the proposal uses the native decoder for the complete regenerated suffix,

$$q_d(\tau' \mid \tau) = P(v' \mid u), \qquad q_d(\tau \mid \tau') = P(v \mid u). \tag{18}$$

The MH ratio at tempering level k is therefore

$$\frac{P_{\eta_k}(\tau')\rho(d)q_d(\tau \mid \tau')}{P_{\eta_k}(\tau)\rho(d)q_d(\tau' \mid \tau)} = \frac{P(u)P(v' \mid u)e^{\eta_k\phi(x_0')}\rho(d)P(v \mid u)}{P(u)P(v \mid u)e^{\eta_k\phi(x_0)}\rho(d)P(v' \mid u)} = e^{\eta_k[\phi(x_0')-\phi(x_0)]}. \tag{19}$$

This proves Eq. (11) and detailed balance with respect to P_{η_k}. Decoder transition probabilities are used to sample the suffix but need not be evaluated: their cancellation is analytical. If the rollback law depended on the current trajectory, its forward and reverse selection probabilities would have to remain in the acceptance ratio.

D Consistency

Under standard conditions for SMC samplers, each reverse-SNIS component $\widehat{p}_{\lambda_j}$ is asymptotically unbiased, although generally biased at a finite particle count [Chopin, 2004, Del Moral et al., 2006]. Our final estimator is a fixed finite convex combination of these component estimators and is therefore also asymptotically unbiased. Independence between components is not required for this conclusion.

E FK Steering implementation

For each terminal tilt λ, choose denoising checkpoints $t_0 < \cdots < t_K$ and a nondecreasing schedule $0 = \beta_0 \le \cdots \le \beta_K = \lambda$. Each particle stores its current partially denoised state, normalized weight, genealogy, and previous potential H. At an intermediate checkpoint, the implementation draws four disposable one-shot texts from the model's predicted clean-token distribution and uses

$$\widetilde{\phi}_k = \log\left(\frac{1}{4}\sum_{s=1}^{4} e^{\phi(\widetilde{x}_{0,k}^{(s)})}\right), \qquad H_k = \beta_k\widetilde{\phi}_k. \tag{20}$$

At the terminal checkpoint it scores the actual stochastic completion and sets $H_K = \lambda\phi(x_0)$. The incremental potential is $G_k = e^{H_k - H_{k-1}}$. Because the stored previous potential is copied during resampling, the product along a lineage telescopes to $e^{\lambda\phi(x_0)}$.

The implementation propagates particles with the native stochastic reverse kernel, evaluates the auxiliary one-shot texts at reward checkpoints, reweights by G_k, and resamples when normalized ESS is below 0.5. The one-shot distribution need not equal the distribution obtained by composing all remaining native denoising transitions. It is used only as an auxiliary look-ahead: the one-shot texts never replace the native propagated particle, and telescoping makes the final target depend on the actual completion. The final weighted population is corrected by Eq. (8), without final resampling.

F Reproducibility details

Tables 2 and 3 summarize the immutable configuration used for the reported results. Every generated artifact records the resolved configuration and is checked against the decoder and evaluator identifiers in the final analysis manifest. All threshold-specific estimates are computed from stored continuous classifier scores, so changing the event threshold does not regenerate model completions. Revision and identifier hashes are abbreviated in the table; the resolved configurations and manifests retain their complete values.

Table 2: Model, evaluator, and estimand configuration. Revisions identify the public snapshots used by the experiment.

Item	Configuration
MDLM checkpoint	`kuleshov-group/mdlm-owt`, revision `d0958fa85133`.
Model implementation	`TheQweaker/mdlm-owt-noflash`, revision `8871c28982b8`; byte-identical official weights with mathematical SDPA.
Tokenizer	`openai-community/gpt2`, revision `607a30d783df`.
Native generation	Unconditional all-mask initial state; length 128; 32 stochastic DDPM/SUBS steps; log-linear noise schedule; temperature one; no top-k or top-p truncation; stochastic categorical final noise removal.
Toxicity evaluator	`SkolkovoInstitute/roberta_toxicity_classifier`, revision `048c25bb1e19`; positive-class index one, with fixed-length padding and padded final microbatches.
Observable and event	$\phi(x) = \log S_{\mathrm{tox}}(x)$ on the first 100 generated tokens; event $S_{\mathrm{tox}}(x) \geq a$ for $a \in \{0.5, 0.9, 0.99, 0.999\}$, with 0.99 primary.
Manifest validation	Decoder ID prefix `cb78b724cdaa`; evaluator ID prefix `91e750abf0a4`; full SHA-256 identifiers stored in every final analysis manifest.

Table 3: Sampling and analysis configuration. Components use independent seeds and are combined with weights fixed before final evaluation.

Item	Configuration
Multi-tilt estimator	$\lambda \in \{0.25, 0.5, 0.75, 1.0\}$; four independent populations; uniform coefficients; no final resampling.
Common particle settings	$N \in \{192, 512\}$; systematic resampling when $\mathrm{ESS}(W)/N < 0.5$; classifier microbatch 16.
FK Steering	Reward checkpoint every four denoising steps; four disposable one-shot clean-token predictions per checkpoint; terminal potential evaluated on the actual native completion.
Trajectory SMC	Eight tempering levels; one MH sweep at every level; rollback depth uniform on $\{4, 8, 16, 24, 32\}$ native steps.
Independent cohorts	Submission: 32 runs at $N = 512$, base seed 500,000,000. Runtime-matched: 32 Trajectory SMC runs at $N = 192$, base seed 900,000,000. Confirmatory: 32 runs per method and budget, base seed 1,100,000,000, budget stride 50,000,000, and disjoint method offsets.
Direct reference	200,000 IID native completions; base seed 8128; append-only shards of 512; generation microbatch 16.
Statistical analysis	Uniform component recombination; 20,000 run-level bootstrap replicates with seed 20260820; primary comparison at $a = 0.99$; $a = 0.999$ exploratory.
Compute environment	One NVIDIA Quadro RTX 6000 per GPU job; Python 3.11.5, PyTorch 2.6.0 with CUDA 11.8, Transformers 4.57.6.

G Additional results

The offline component export contains 3,584 rows spanning four thresholds, four tilts, and all independent runs. Uniform recombination agrees with the stored final estimates to machine precision. Figure 2 contrasts genealogy and exact terminal-token duplication. Markers and bands show medians and interquartile ranges over 64 independent $N = 512$ populations.

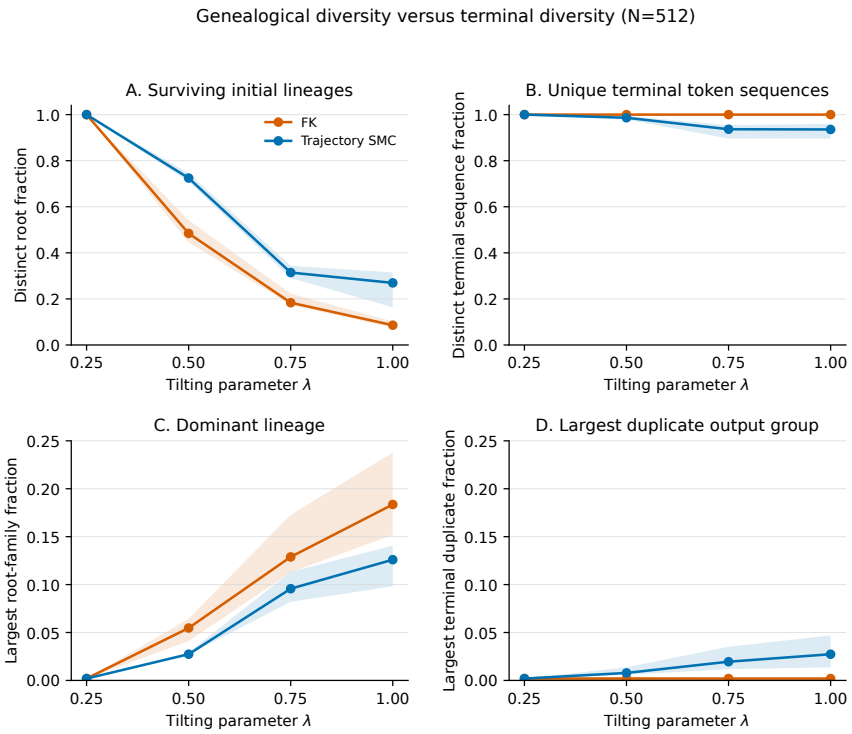


Figure 2: Genealogical and terminal-sequence diversity across tilts. Surviving root lineages and dominant root-family size measure dependence created by resampling; exact terminal-sequence uniqueness and the largest duplicate group measure realized token-level duplication. Their different orderings show that the two notions must be reported separately.